Chapter 26

Quality Assurance in Pharmacy Education

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Learning Objectives

Objective 26.1 Describe quality assurance of pharmacy curricula in a global perspective.
Objective 26.2 Explain the role of ACPE in accreditation and certification of pharmacy programs.
Objective 26.3 Detail the curriculum mapping process in pharmacy education.

OBJECTIVE 26.1. DESCRIBE QUALITY ASSURANCE OF PHARMACY CURRICULA IN A GLOBAL PERSPECTIVE

Responsibilities of pharmacists are evolving globally to meet public health demands. Once involved primarily with dispensing and compounding medications, pharmacists worldwide are increasingly serving as medication experts working as members of interprofessional healthcare teams.[1] Within individual countries, pharmacy curricula must adapt to address the changing nature of responsibilities for the country's pharmacy workforce as the ability to provide optimal pharmaceutical care depends on the existence of a qualified and competent pharmacy workforce. Governments and academic institutions increasingly need to reevaluate pharmacy education and the methods used to measure, assure, and improve academic quality. Recognizing the need to advance pharmacy education and the quality assurance process, the International Pharmaceutical Federation (FIP), in conjunction with the World Health Organization (WHO) and the United Nations Educational, Scientific and Cultural Organization launched the Pharmacy Education Taskforce (PET) in 2007 to oversee implementation of the 2008–10 Pharmacy Education Action Plan.[2] As defined by PET, "*pharmacy education refers to the educational design and capacity to develop the workforce for a diversity of settings (e.g., community, hospital, research and development, academia) across varying levels of service provision and competence (e.g., technical support, human resources, pharmacist practitioners, pharmaceutical scientists, pre-service students) and scope of education (e.g., undergraduate, post-registration, continuing professional development, practitioner development, life-long learning).*"[3]

Pharmacy Education Planning

The Pharmacy Education Action Plan (hereafter referred to as the Action Plan) advocates for a needs-based approach to pharmacy education (see Fig. 26.1).[1] Central to the Needs-Based Education Model is the vision for pharmacy practice and education, which should be developed through profession-wide consensus. The Action Plan encourages the development of educational systems designed to produce a pharmacy workforce competent to deliver healthcare services that are aligned with the local needs of the country. The Action Plan provides evidence-based guidance and frameworks that can be used to advance pharmacy education and encompasses four domains of action including: (1) development of a vision and framework for education development; (2) development of a competency framework; (3) ensuring a quality assurance system; and (4) building academic and institution capacity.

Within healthcare, the quality assurance process supports and facilitates improvements in practice because educational institutions are expected to modify the education provided in an effort to address advances in knowledge and practice.

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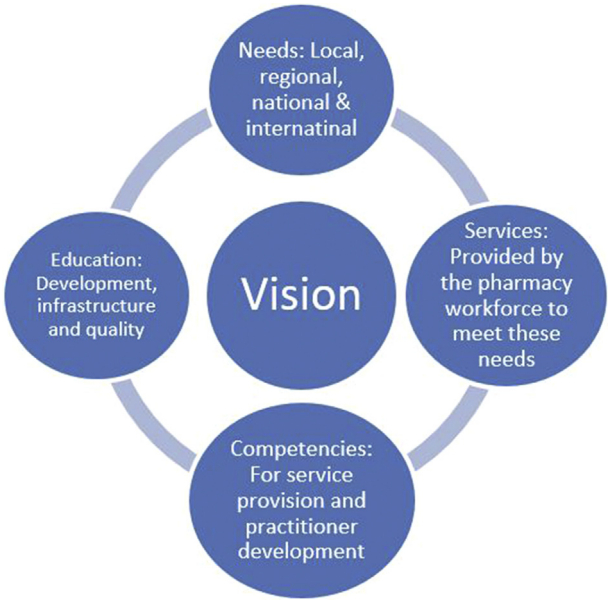


FIGURE 26.1 FIP education needs-based education model *Adapted from FIP, Quality Assurance of Pharmacy Education: The FIP Global Framework. 2nd Edition; 2014.*

Framework-Based Quality Assurance Systems Development

Quality assurance systems vary greatly between countries. Although many countries have well-developed quality assurance systems that utilize standards aligned to local healthcare needs, in other countries such systems may be lacking or still emerging. Ideally, each country should have its own national system for quality assurance and standards for pharmacy education that reflect contemporary pharmacy practice and education and meet the country's needs.[4] As the principles and core elements utilized in the quality assurance process are universal, countries seeking to establish or improve their system could utilize an internationally developed and adopted framework for quality assurance of pharmacy education for reference purposes.

The FIP International Forum for Quality Assurance of Pharmacy Education developed a globally applicable framework for quality assurance of pharmacy education that incorporates core principles and elements considered essential for an effective approach to quality assurance.[4]

FIP's Quality Assurance Framework (Framework) provides a template that can be modified to suit local needs and conditions within individual countries and focuses on the elements that need to be included and applied. The Framework addresses three main sections. The *first section*, Prerequisites for Quality Assurance in Pharmacy Education, outlines the prerequisites and foundational elements that need to be identified and will form the basis of the design, development, and implementation of the quality assurance system. For example, individual countries would need to identify the national vision for pharmacy practice and education within their country. Essential to this vision will be the identification of national, societal, and population healthcare needs that form the basis for the pharmacist's responsibilities within the healthcare system. To ensure the vision adequately encompasses the full scope of healthcare needs, a broad array of relevant stakeholders should be identified and engaged in the process. The core healthcare needs and issues of countries are likely to be the same or similar, but differences will certainly exist. These differences are the result of or are influenced by several factors, including population demographics, public health, and morbidity challenges and trends, and national economic and political priorities.

The *second section*, Quality Criteria and Quality Indicators for Pharmacy Education, outlines the criteria that should be addressed to ensure quality. These criteria include the five "pillars" of quality (context, structure, process, outcomes, and impact) of pharmacy education as well as the three "foundations" of competency that should be addressed in pharmacy education (science, practice, and ethics) (see Fig. 26.2).

Once the vision for pharmacy education and practice has been established, the philosophy and purpose of quality assurance can be addressed. The third section of the framework addresses components that should be considered during development of the quality assurance system. An initial step involves establishing the structure, mandate, and purpose of the quality assurance agency. Different models for quality assurance can be considered with the entity charged with

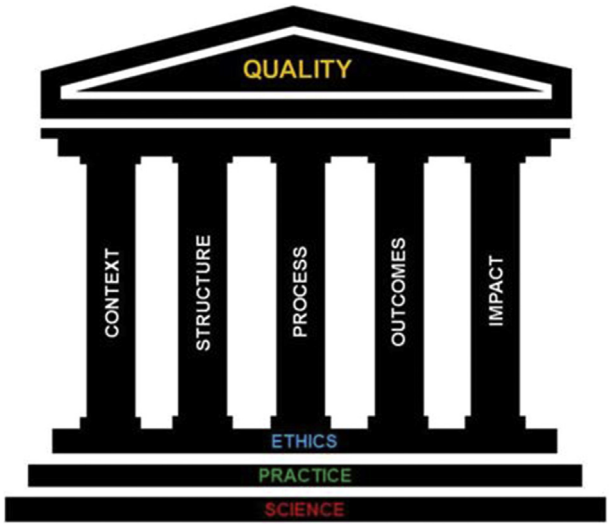


FIGURE 26.2 The pillars and foundations of educational quality *Adapted from FIP, Quality Assurance of Pharmacy Education: The FIP Global Framework. 2nd Edition; 2014.*

oversight of the quality assurance process falling under a governmental agency, a national pharmacy organization, or an independent, autonomous agency. The level of quality assurance, either at the institutional or program-level, should also be established. The composition of the agency's governing/decision-making board or council should be identified, including the criteria for appointment or selection of members and the duration of service. Identification of funding sources to support the agency is essential to ensuring the sustainability of its operations. Policies and procedure need to be established to ensure consistent and fair accreditation decisions that are free from any conflict of interest. The procedures that will be used to evaluate individual pharmacy education programs should form the basis for the decision-making process. The criteria/standards should be clearly articulated, evidence-based, validated, publicly disclosed, and endorsed profession-wide. Periodic updating of the criteria/standards is needed to ensure continued applicability in the face of changes in practice and changing national healthcare needs.

OBJECTIVE 26.2. EXPLAIN THE ROLE OF ACPE IN ACCREDITATION AND CERTIFICATION OF PHARMACY PROGRAMS

Structure and Mission

In most countries, the national government has the responsibility—directly or indirectly—for quality assurance of higher education, including professional degree programs, such as pharmacy. A growing trend, however, is the establishment of independent, autonomous agencies that have a specialized- or programmatic focus of activity. One such example is the Accreditation Council for Pharmacy Education (ACPE), the national agency for the accreditation of professional degree programs in pharmacy and providers of continuing pharmacy education in the United States. ACPE (until 2003 known as the American Council on Pharmaceutical Education) was established in 1932 for the accreditation of professional degree programs in pharmacy, and its scope was expanded in 1975 to include accreditation of providers of continuing pharmacy education (www.acpe-accredit.org). ACPE further broadened its activities to include evaluation and certification of professional degree programs internationally in 2011 and entered into a collaboration with the American Society of Health-System Pharmacists to accredit pharmacy technician education and training programs beginning in 2014. ACPE is a not-for-profit agency, whose mission and sole purpose are to assure and advance quality in pharmacy education.

To address its mission, ACPE routinely interacts with a variety of stakeholders. The 10 members of ACPE's Board of Directors are appointed by the American Association of Colleges of Pharmacy (AACP), the American Pharmacists Association (APhA), the National Association of Boards of Pharmacy (NABP) (three members each), and the American Council on Education (ACE) (one nonpharmacy-related appointment). This structure ensures that the Board includes perspectives and experience from regulators (NABP appointees), practitioners (APhA appointees), and educators (AACP appointees) as well as the public (ACE appointee). In a broader context, ACPE routinely interacts with

stakeholders to ensure the organization continues to adapt to the changes within the pharmacy profession in the United States and globally. Engagement with state boards of pharmacy, state and federal governments, and other accreditation agencies ensures ACPE remains informed of regulatory and policy changes. Similarly, changes in education are captured through interactions with pharmacy programs, individual educators, and continuing education providers, whereas changes in practice are identified through interactions with pharmacy practitioner organizations, employers, other health professionals, students, and the public. ACPE works with organizations outside of the profession. In this regard, ACPE, along with several other health disciplines accreditation agencies, formed the Health Professions Accreditors Collaborative to strengthen communication between the healthcare professions and enhance interprofessional education and care. In addition, ACPE, in conjunction with the Accreditation Council for Continuing Medical Education (ACCME) and the American Nurses Credentialing Center (ANCC) provide a process for Joint Accreditation for Interprofessional Continuing Education providers. The joint accreditation process offers providers the opportunity for accreditation by all three providers (ACPE, ACCME, and ANCC) through a single review process and promotes interprofessional continuing education activities that are designed "by the team, for the team" to enhance interprofessional collaborative practice.

US state boards of pharmacy require that domestic licensure applicants be a graduate of an ACPE-accredited pharmacy degree program to be eligible to sit for the North American Pharmacist Licensure Examination (NAPLEX). Accreditation provides a national basis for quality assurance of the education students receive. The value of the accreditation process is multifactorial, serving multiple stakeholders, including the public, students and prospective students, licensing bodies, pharmacy programs and their parent institutions, and the profession. ACPE accreditation provides public recognition that a professional degree program has met the required standards through a series of initial and subsequent periodic evaluations.[5] Graduates of accredited programs meet the educational requirements for licensure and should be educationally prepared for practice. ACPE standards are minimum requirements; programs commonly exceed the minimum requirements by using a continuous quality improvement process. The standards, based on evidence and experience, describe the various elements needed for quality-assured professional education. The standards describe expectations that ACPE has of academic institutions offering the Doctor of Pharmacy degree program. ACPE standards, policies, and procedures also address the expectations that the US Department of Education and state boards of pharmacy have for the quality assurance of professional degree programs.

Many countries have rigorous systems and requirements in place that make it possible for foreign graduates to achieve licensure to practice. For example, in the United States, graduates of foreign pharmacy programs that are not accredited by ACPE are required to pass the Foreign Pharmacy Graduate Equivalency Examination (FPGEE) and TOEFL iBT, an English-language exam, before they are eligible to sit for the license exam. The education of foreign and US graduates may not be equivalent per se, but the evaluation (quality assurance) process—known as the Foreign Pharmacy Graduate Examination Committee Certification (FPGEC)—ensures that a foreign pharmacist's education meets acceptable requirements to allow them to practice as licensed pharmacists.

Standards Development

ACPE's standards are developed with input from a broad range of stakeholders interested in and affected by pharmacy education. The ACPE standards focus on the educational outcomes required of Doctor of Pharmacy programs and the assessment of those outcomes. ACPE's current standards specifically target the Center for the Advancement of Pharmacy Education (CAPE) Educational Outcomes.[6] To promote consistency in care, it is expected that the curriculum will prepare students to provide patient-centered collaborative care as noted in the Joint Commission of Pharmacy Practitioners' Pharmacists' Patient Care Process model.[7] The structural and process-related elements within pharmacy education that are necessary to implement evidence-based outcomes are described within the standards to assist programs with documenting achievement of the standards. In addition, the standards describe areas where programs can experiment and innovate within the didactic and experiential components of their curricula to meet the required educational outcomes. Establishing a commitment to life-long learning and the transition to continuing professional development (CPD) by students and graduates is addressed within the standards, which require that programs impart to graduates the knowledge, skills, abilities, behaviors, and attitudes necessary to demonstrate self-awareness, leadership, innovation, entrepreneurship, and professionalism. The ACPE standards expect that the program will utilize a plan to assess student readiness to enter final year pharmacy practice experiences, provide direct patient care in a variety of healthcare settings ("practice-ready"), and contribute as a member of an interprofessional collaborative patient care team ("team-ready"). To provide evidence of achievement of the standards, programs must provide, at a minimum, documentation related to student achievement throughout the program including: (1) student progression rates; and (2) student performance on standardized examinations both during the didactic curriculum and on the national licensure and state law examinations.

Accreditation Process

The accreditation process is designed to provide a professional judgment as to the quality of a college or school of pharmacy's professional program and to encourage continued improvement thereof. Accreditation concerns itself with both quality assurance and quality enhancement. Recognition does not imply or infer that all Doctor of Pharmacy programs are equivalent beyond meeting the expectations of the accreditation standards. Through the accreditation process, a professional program is evaluated on the extent to which it accomplishes its stated goals and is consistent with the concept that pharmacy is a unique, personal service profession in the health science field. In the application of these standards, literal conformity in every detail is not required beyond meeting the minimum expectations of the standards. The accreditation process assures basic expectations for quality pharmacy education. Many programs exceed the standards in one or more of the various elements.

The accreditation process is based on a structured and systematic system of self and peer assessment with the intention of improving academic quality and providing accountability. In this regard, the accreditation process includes an in-depth self-evaluation ("self-study") conducted by the program seeking initial or continued accreditation. A comprehensive on-site evaluation of the program is conducted by a team of trained evaluators selected by ACPE, the members of which have diverse academic and practice backgrounds and experience. The evaluation team reviews the program's self-study report that includes required documentation and data, and visits the program to evaluate the program's compliance with the standards. The evaluation team provides a written report of its findings and recommendations, which are reviewed by the ACPE Board of Directors. The Board is the decision-making entity and makes all accreditation determinations. Each professional degree program is reviewed comprehensively every 8 years. The Board may act to require additional monitoring, through focused on-site evaluations and/or written interim reports, during the standard accreditation period for any standards that are not adequately addressed. Alternatively, the Board may award a shortened term of accreditation to programs with significant compliance concerns. Programs that fail to address compliance concerns of the Board in a timely fashion may be placed on Probation (which is a public notification) and continued failure to address the issue will result in withdrawal of accreditation; such decisions are, however, subject to appeal by the institution.

ACPE's International Services Program (ISP) offers consultation, training, and degree program evaluation services to stakeholders around the world who seek guidance and assistance related to quality assurance and advancement of pharmacy education. The expertise, global perspectives, staff resources, and formal processes within the ISP support international stakeholders to advance pharmacy education and quality in their respective countries. The activities and programs of the ISP are supported and guided by the International Commission whose international and US-based members are appointed by the ACPE Board of Directors, and the ISP Advisory Group whose expert members are drawn from around the world.

OBJECTIVE 26.3. DETAIL THE CURRICULUM MAPPING PROCESS IN PHARMACY EDUCATION

The role of pharmacists within healthcare is changing and pharmacy curricula need to adapt to these changes as well as changing societal needs and priorities, and evolving quality assurance standards/criteria. To ensure continued relevance and applicability of the curriculum, pharmacy programs must have a clearly defined process for evaluating the curriculum. Typically, the curriculum committee is the entity charged with curricular oversight including ensuring the curriculum aligns with both programmatic outcomes and national quality assurance standards.

A program's curriculum can be viewed through three different lenses: (1) the intended or designed curriculum; (2) the delivered or taught curriculum; and (3) the experienced or learned curriculum.[8] The "intended" curriculum is the curriculum captured on paper. It is the formal, documented curriculum including the course titles, written course content, individual learning objectives, and assessment measures. The "delivered" curriculum encompasses the content delivered by the faculty including the teaching method(s) used in delivery. The "experienced" curriculum describes the curriculum as experienced by the learners. Ideally, all three, the *intended*, *delivered*, and *experienced* will be aligned.

Curriculum mapping is a technique used to document how curricular content, teaching methods, and assessments are incorporated throughout the pharmacy curriculum in an effort to ensure achievement of programmatic objectives, i.e., competencies of the graduates. The curriculum mapping process evaluates when, how, and what is taught within the curriculum and considers the assessment measures used to evaluate achievement of expected student learning outcomes.[9] Optimally curricular mapping is an ongoing process intended to provide a real-time picture of the curriculum, not a single-point-in-time initiative.

The curricular mapping process provides a means to evaluate the curriculum and interpret curricular data. Curriculum mapping within pharmacy education can be used to identify curricular gaps or excessive redundancies in content to ensure

TABLE 26.1 Example Curricular Mapping Matrix

Intended Student Learning Outcomes	Course A	Course B	Course C	Course D
Outcome 1	Introduced		Reinforced	Mastered
Outcome 2		Introduced		
Outcome 3		Introduced	Reinforced	Reinforced

all programmatic outcomes are adequately addressed throughout the course of the curriculum. Curriculum mapping can also assist in planning for assessment methods to evaluate the curriculum.[10]

The curricular mapping process looks at the intended/delivered curriculum as well as the experienced curriculum. Mapping of course syllabi and the learning objectives for both individual courses and lectures can be used to identify where, when, and how in the curriculum course content is located and delivered. The mapping process should take into consideration the breadth and depth of exposure to curricular content (introduction, reinforcement, and mastery of curricular content addressing knowledge, skills, attitudes, and values) and the type of teaching methods used in the delivery of the content (e.g., passive/lecture based delivery, active learning method, application-based activities) (see Table 26.1: Example Curricular Mapping Matrix). The experienced curriculum should also be mapped using assessment methods including multiple choice examinations, student surveys, Objective Structured Clinical Examinations (OSCEs), and review of student portfolios.

Software programs (e.g., ExamSoft) including electronic testing software can be used to assist in the curricular mapping process in an effort to facilitate data collection and analysis. The ability to search the database using key terms and concepts is essential to ensuring the usability of data gathered. In addition, the database should be accessible to facilitate real-time, dynamic reporting.

Data gathered through the curricular mapping process should be used to document relationships between the curriculum and the intended learning objectives. Analysis of data collected should ensure that the curriculum adequately addresses each of the programmatic outcomes. Areas where gaps or redundancies exist should be reviewed by the curriculum committee. Additionally, the curricular mapping process can be used to identify opportunities for curricular integration. Ideally, the curricular map should facilitate communication among faculty and provide the basis for the identification of needed curricular revisions and continuous quality improvement.

CONCLUSION

Pharmacy education must adapt to the changing needs of the profession. Quality assurance procedures help to assure and advance the quality of pharmacy education. It is essential, however, that all sectors within the profession—primarily those representing education, practice, policy, and regulation communicate and collaborate effectively to advance the profession in line with a common vision.[11] Quality assurance agencies—whether governmental or independent—should likewise collaborate with all stakeholders. The competencies that graduates need for practice must be aligned with the educational outcomes of pharmacy degree programs required by quality assurance agencies, and further aligned with blueprints of national licensure examinations. Licensure examinations must be rigorous and psychometrically valid to ensure that competencies of graduates are appropriately and accurately assessed. Such examinations can also serve as useful indicators of programmatic quality and curricular effectiveness and provide useful data for quality improvement.

The Pharmacy Education Action Plan, developed by FIP's Pharmacy Education Taskforce, encourages the development of educational systems designed to ensure the competency of the pharmacy workforce. Quality assurance systems, such as that utilized by the Accreditation Council for Pharmacy Education in the United States and other countries, are designed to provide assurance of the quality of professional degree programs, to the profession and the public. Individual programs can advance the quality of education provided through continuous quality improvement initiatives such as curricular mapping.